

SAIM HAIDER

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PROFESSIONAL SUMMARY

Recent BSCS graduate from FAST-NUCES with hands-on experience in AI/ML model fine-tuning, backend systems engineering, and containerized application development. Built a production-grade AI pipeline that fine-tuned CodeLlama using LoRA to generate verified SDN scripts with zero execution errors. Strong foundation in Python, Django, and Docker with practical experience shipping event-driven microservices. Eager to contribute to a real engineering team while growing in a fast-paced environment.

EDUCATION

BS Computer Science | FAST-NUCES, Peshawar | Sep 2022 – Jun 2026

- Relevant coursework: Data Structures & Algorithms, OOP, Database Systems, Operating Systems, AI

WORK EXPERIENCE

Teaching Assistant — Technical & Business Writing | FAST-NUCES, Peshawar | 2024 – 2025

- Mentored CS students in structuring professional technical documentation, business proposals, and corporate communications
- Evaluated assignments and delivered constructive written and verbal feedback to a class of 30+ students
- Assisted faculty in managing classroom activities, demonstrating reliability and leadership in an academic setting

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript (ES6+), HTML/CSS

Frameworks & Libraries: Django, React.js

DevOps & Tools: Docker, Docker Compose, Git/GitHub, Linux/Bash

AI/ML: LLM fine-tuning, LoRA (PEFT), CodeLlama, Hugging Face, Google Colab, Streamlit

Databases: MySQL, PostgreSQL, SQLite

Concepts: OOP, REST APIs, Microservices, Event-Driven Architecture, MVC, WSGI

PROJECTS

ScriptNet Edu — AI-Driven SDN Script Generator — *Python · CodeLlama · LoRA · Mininet · FastAPI · Streamlit*

Fine-tuned CodeLlama to generate verified Python scripts for Mininet SDN simulations from natural language — eliminating hallucination errors common in general-purpose LLMs.

- Fine-tuned CodeLlama-7B using LoRA (Low-Rank Adaptation) on a custom Mininet/SDN dataset — achieving domain-specific syntax accuracy without full retraining
- Engineered a defensive validation backend that intercepted every generated script and ran automated execution tests before returning output — achieving zero execution errors on targeted topologies
- Exposed the full pipeline through structured API endpoints handling prompt input, token authorization, and verified JSON script responses
- Applied 4-bit quantization to optimize model inference on consumer hardware without sacrificing output quality

Event-Driven E-Commerce Microservices — *Python · Django · Docker · RabbitMQ · MySQL*

- Architected a decoupled e-commerce backend separating Admin and User services, connected via RabbitMQ message broker for asynchronous data consistency
- Containerized the full stack — Django, MySQL, RabbitMQ — using Docker Compose with internal networking, persistent volumes, and health checks
- Implemented retry logic and Safe Mode error handling to prevent service crashes during race conditions or broker downtime

Custom WSGI-Compliant Web Server — *Python · Sockets · Django · Flask*

- Built a functional HTTP/1.1 web server from scratch using Python socket programming, managing raw TCP/IP connections without high-level abstractions
- Implemented the PEP 333 WSGI interface to successfully deploy both Django and Flask applications on the custom server

Enterprise DOM-Bypass Chrome Extension — *React.js · JavaScript ES6+ · Chrome APIs*

- Built a React-based Chrome extension that injects dynamic UI components and bypasses enterprise Virtual DOM rendering constraints
- Managed complex component state and lifecycle within a sandboxed browser environment to deliver a low-latency user experience

ADDITIONAL

Languages: English (Professional), Urdu (Native)